S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR

Practical No. 07

Aim: Write a Prolog Program to demonstrate the concept of Family relationship.

Name of Student:

Roll No.: Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: Write a Prolog Program to demonstrate the concept of Family relationship.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- Represent knowledge using **facts**
- Define rules using **logical inference**
- Understand **unification**
- Observe **backtracking**
- Model **hierarchical relationships** (parent, sibling, grandparent, etc.)

THEORY:

Prolog (Programming in Logic) is a **declarative logic programming language** based on:

- First-Order Predicate Logic
- Resolution
- Unification
- Backtracking

Unlike procedural languages (C, Python), Prolog focuses on **what is true**, not **how to compute it**.
Common implementation: **SWI-Prolog**
In Prolog a program consists of:

1. Facts
2. Rules
3. Queries

Installing Prolog (Using SWI-Prolog)

Step 1: Download

Visit: https://www.swi-prolog.org

Step 2: Install

Follow OS-specific installer.

Step 3: Run

Open terminal / command prompt:

swipl

?- This is the Prolog query prompt.

Prolog data types:

- **Atoms** → `john`, `apple`
- **Variables** → `X`, `Person`
- **Numbers** → `10`, `3.14`
- **Structures** → `parent(john, mary)`
- **Lists** → `[1,2,3]`

Unification:

Prolog matches patterns for unification. Exa:

parent (john, X) Unifies with: parent (john, mary)

Result:

```
X = mary
```

Backtracking:

Prolog automatically searches for alternatives using backtracking.
Example Query:

?- parent (john, X).

Press ; to get next solution.

Important Operators:

Operator	Meaning
`:-`	If
`,`	AND
`;`	OR
`=`	Unification
`is`	Arithmetic evaluation
`\=`	Not unifiable
`\+`	Negation as failure

A **family relationship system** is a classical example used in Artificial Intelligence to demonstrate:

- Knowledge representation
- Predicate logic
- Recursive reasoning
- Depth-first search with backtracking

ALGORITHM:

1. Start
2. Define family members using `male/1` and `female/1` facts
3. Define parent relationships

4. Define derived relationships using rules:

    a. father
    b. mother
    c. sibling
    d. grandparent
    e. ancestor

5. Load the program in Prolog
6. Execute queries
7. Observe results and backtracking
8. Stop

PROGRAM CODE:

Create a file named: `family.pl`

```
% ---------- Facts ----------

male(gunwant).
male(jiwandhar).
male(rajendra).
male(rohit).
male(rohan).

female(shantidevi).
female(padma).
female(sangita).

parent(gunwant, jiwandhar).
parent(shantidevi, jiwandhar).

parent(jiwandhar, rajendra).
parent(padma, rajendra).

parent(rajendra, rohit).
parent(sangita, rohit).

parent(rajendra, rohan).
parent(sangita, rohan).

% ---------- Rules ----------

father(X, Y) :-
    parent(X, Y),
    male(X).
```

```
mother(X, Y) :-
    parent(X, Y),
    female(X).

sibling(X, Y) :-
    parent(Z, X),
    parent(Z, Y),
    X \= Y.

grandparent(X, Y) :-
    parent(X, Z),
    parent(Z, Y).

ancestor(X, Y) :-
    parent(X, Y).

ancestor(X, Y) :-
    parent(X, Z),
    ancestor(Z, Y).
```

INPUT & OUTPUT:

Step 1: Start Prolog

```
swipl
```

Step 2: Consult File

```
?- consult('family.pl').
```

OR

```
?- [family].
```

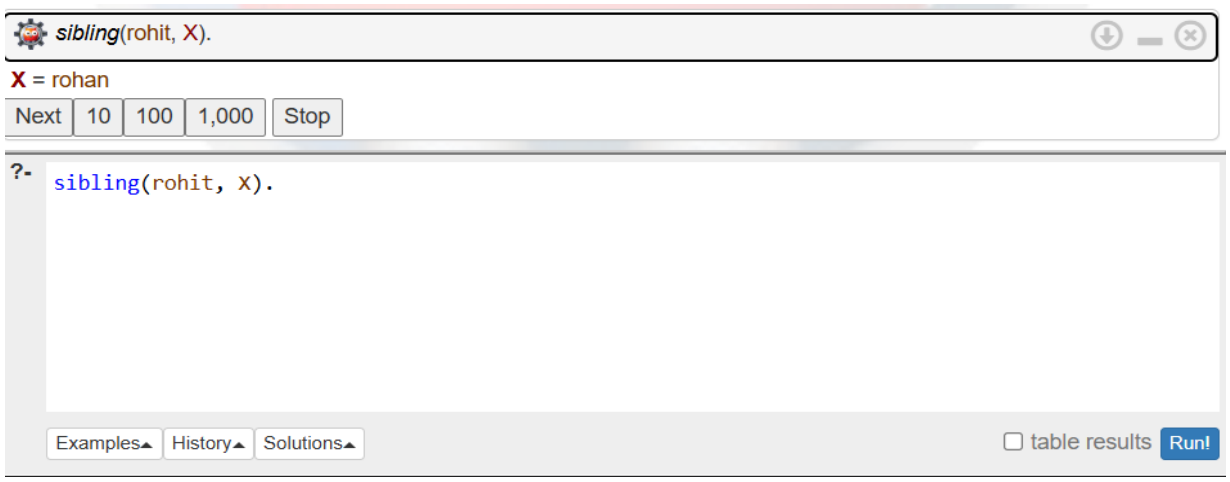
Queries & Output

1. Find father of Sam

```
?- father(X, sam).
```

2. Find siblings of rohit

```
?- sibling(rohit, X).
```

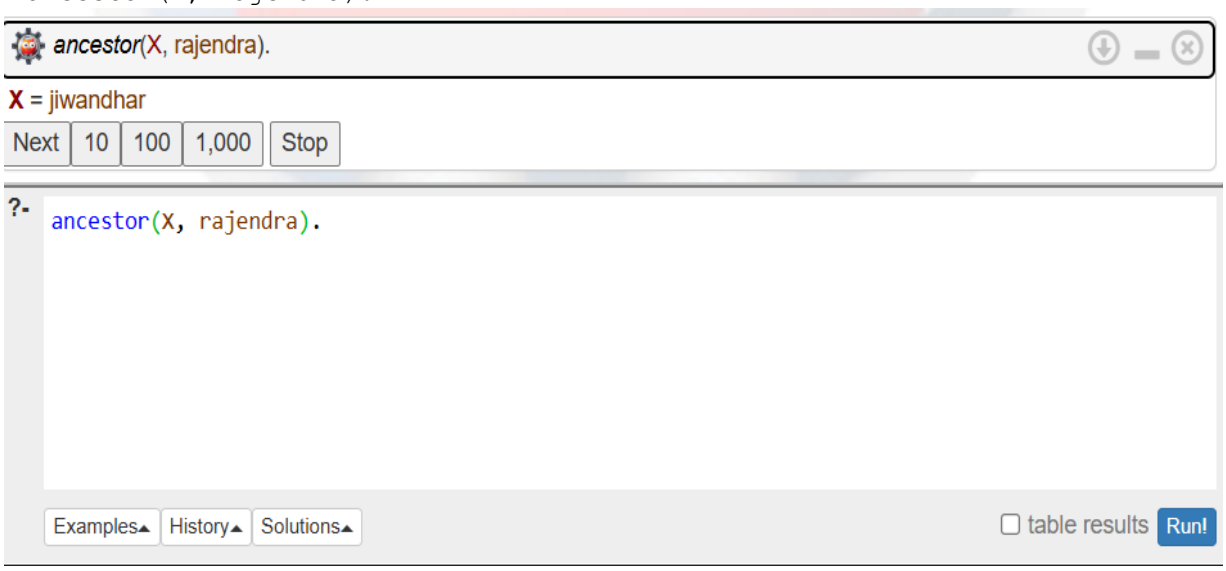


3.Find grandparents of rohit

```
?- grandparent(X, rohit).
```

4.Find ancestors of rajendra

```
?- ancestor(X, rajendra).
```



CONCLUSION:

DISCUSSION QUESTIONS:

1. What is a fact in Prolog?

2. What is the difference between fact and rule?

3. What is unification?

4. Explain backtracking with example.

5. Why is recursion used in `ancestor` rule?

6. What is the difference between `=` and `is`?

REFERENCES:

- https://www.swi-prolog.org/
- https://www.tutorialspoint.com/prolog/prolog_introduction.htm
- https://www.geeksforgeeks.org/artificial-intelligence/prolog-an-introduction/
- https://www.youtube.com/watch?v=dKn-BbS_zQQ